Molecular Dynamics Insights into Steroid Modulation of NMDA Receptors

Yuan Chang-Halabi[1,2], Marcel Bermúdez[1,2]

[1] Institute of Pharmaceutical and Medicinal Chemistry, Universität Münster; [2] GRK 2515, Chemical Biology of Ion Channels (Chembion), Universität Münster

(corresponding author: yuan.chang@uni-muenster.de)

Keywords: NMDA receptor, neurosteroids, MD simulations, Allosteric modulation

ABSTRACT

Ion channels are embedded in complex lipid environments that modulate their structure, dynamics, and function[1]. The membrane-mediated modulation of the N-methyl-D-aspartate receptor (NMDAR) has been recognized for decades; however, the underlying molecular mechanisms are not fully understood[2,3]. This tetrameric glutamate-gated ion channel has been implicated in numerous neurological and neuropsychiatric disorders, including Alzheimer's disease and depression, while also posing significant challenges for the development of safe and selective therapeutics[4]. Positive allosteric modulation of NMDARs has been proposed to involve specific receptor–steroid interactions with endogenous molecules such as pregnenolone sulfate and cholesterol, as well as synthetic steroid derivatives. Using molecular dynamics simulations, dynamic pharmacophore modelling, and allosteric network analysis, we aim to characterize the molecular determinants of steroid recognition and elucidate the mechanisms underlying positive allosteric modulation of NMDARs. This membrane-centric approach may facilitate the rational design of novel allosteric modulators of NMDARs.

REFERENCES

1 *Harayama, T.; Riezman, H. Nature Reviews Molecular Cell Biology 2018, 19 (5), 281–296. https://doi.org/10.1038/nrm.2017.138.*

2 *Kang, H. et al. Nature* ***2025****, 648 (8092), 220–228. https://doi.org/10.1038/s41586-025-09695-4*

3 Korinek, M. et al. *The Journal of Physiology* **2015**, *593* (10), 2279–2293. https://doi.org/10.1113/jphysiol.2014.288209

4 Zeng, Y. et al. Acta Pharmacologica Sinica 2025, 47 (1), 3–21. https://doi.org/10.1038/s41401-025-01675-8.